WORKED SOLUTION

STEP III 2025, Question 4

Pure • Transformations of Curves

STEP III 2025, Question 4

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

The key idea

Suppose a map T of the plane has an inverse. Then the curve $f(T^{-1}(x,y)) = 0$ is the image under T of the curve $f(x,y) = 0$, because a point lies on the new curve exactly when T^{-1} sends it to the old one. Part (i) shows this for a rotation. Part (ii) does the same for a reflection, where $T^{-1} = T$. In part (iii) we spot that the substitution $u = x - 2y$ turns the equation into a horizontal shear of $y = \sin x$, and a parametrisation then gives the tangents with no implicit-differentiation headaches.

Solution

Part (i)

Multiplying out the matrix product gives

$$x_2 = \frac{x_1 - y_1}{\sqrt{2}}, \qquad y_2 = \frac{x_1 + y_1}{\sqrt{2}}.$$

Substitute $(x,y) = (x_2, y_2)$ into the two brackets in the equation of G_2:

$$\frac{x_2}{\sqrt{2}} + \frac{y_2}{\sqrt{2}} = \frac{(x_1 - y_1) + (x_1 + y_1)}{2} = x_1,$$
$$-\frac{x_2}{\sqrt{2}} + \frac{y_2}{\sqrt{2}} = \frac{-(x_1 - y_1) + (x_1 + y_1)}{2} = y_1.$$

So the left-hand side of the equation of G_2 at (x_2, y_2) is $\frac{x_1^2}{9} + \frac{y_1^2}{4}$. This equals 1 because (x_1, y_1) is on G_1. Hence (x_2, y_2) lies on G_2.

For the rotation, note that $\frac{1}{\sqrt{2}} = \cos 45° = \sin 45°$, so the matrix is

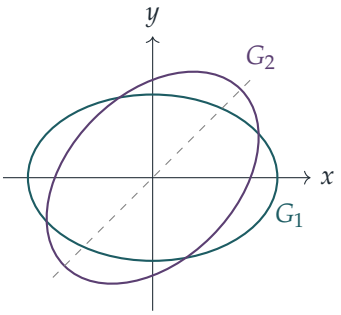
$$R = \begin{pmatrix} \cos 45° & -\sin 45° \\ \sin 45° & \cos 45° \end{pmatrix},$$

the anticlockwise rotation through $45°$ about the origin. We have just shown that R maps every point of G_1 onto G_2. To show that G_2 is *exactly* the rotated curve, we also

need every point of G_2 to come from G_1. Take any (x, y) on G_2 and put

$$x_1 = \frac{x}{\sqrt{2}} + \frac{y}{\sqrt{2}}, \qquad y_1 = -\frac{x}{\sqrt{2}} + \frac{y}{\sqrt{2}},$$

which is $R^{-1}(x, y)$ (rotation back through 45°). The equation of G_2 says precisely that $\frac{x_1^2}{9} + \frac{y_1^2}{4} = 1$, so $(x_1, y_1) \in G_1$, and $R(x_1, y_1) = (x, y)$. Therefore $G_2 = R(G_1)$: the ellipse G_2 is G_1 rotated anticlockwise through 45° about the origin.



Part (ii)(a)

Call the matrix M. A point (x, y) is invariant when $M\binom{x}{y} = \binom{x}{y}$, that is,

$$-0.6x + 0.8y = x, \qquad 0.8x + 0.6y = y.$$

The first gives $0.8y = 1.6x$, so $y = 2x$, and the second gives $0.8x = 0.4y$, which is the same thing. So the line of invariant points (the mirror line) is

$$y = 2x.$$

Part (ii)(b)

Write $(X, Y) = M(x, y) = (-0.6x + 0.8y,\ 0.8x + 0.6y)$. The second equation says exactly

$$Y = 2^X.$$

So (x, y) is on the second graph if and only if $M(x, y)$ is on $y = 2^x$. A reflection undoes itself ($M^2 = I$, as you can check by multiplying out), so $(x, y) = M(X, Y)$. This is the part (i) argument again: the second graph is the reflection of $y = 2^x$ in the line $y = 2x$.

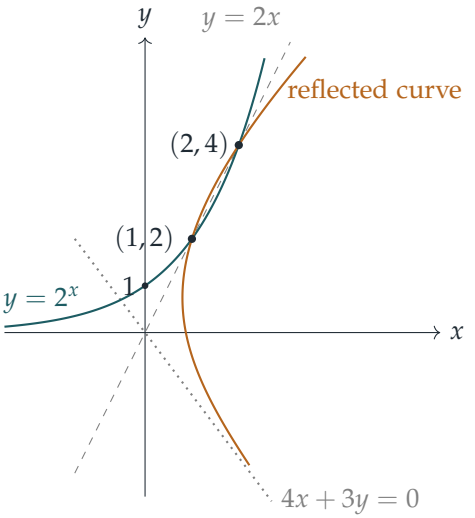
? **Ask yourself** Where does the asymptote of the reflected curve go, and which way does the curve run off to infinity?

Reflect the features of $y = 2^x$ one at a time.

- *Asymptote.* As $x \to -\infty$, $y = 2^x$ approaches the x-axis. The direction $(1, 0)$ reflects to $M(1, 0) = (-0.6, 0.8)$, so the x-axis reflects to the line $4x + 3y = 0$, i.e. $y = -\frac{4}{3}x$. Equivalently, $Y = 0.8x + 0.6y = 2^X > 0$, so the reflected curve lies strictly on the side

$4x + 3y > 0$ and approaches $4x + 3y = 0$ as $X \to -\infty$. Far out, the points of $y = 2^x$ are about $(X, 0)$ with X large and negative, and these reflect to about $(-0.6X, 0.8X)$. So the reflected curve approaches its asymptote in the fourth quadrant, with $x \to +\infty$ and $y \to -\infty$.

- *Points on the mirror.* The reflection fixes every point on $y = 2x$. Now $2^x = 2x$ at $x = 1$ and $x = 2$ (and only there, since $2^x - 2x$ is convex). So both curves pass through $(1, 2)$ and $(2, 4)$, and they cross there. Searching by computer along $y = 2^x$ finds no other common points.
- *Other points.* $(0, 1)$ reflects to $(0.8, 0.6)$.
- *Leftmost point.* The reflected curve is $\left(-0.6t + 0.8 \cdot 2^t,\ 0.8t + 0.6 \cdot 2^t\right)$. Its x-coordinate is least where $0.8 \cdot 2^t \ln 2 = 0.6$, at about $(0.80, 0.74)$. There the tangent is vertical, so the reflected curve is not the graph of a function of x.
- *Growth.* As $X \to +\infty$ the points $(X, 2^X)$ reflect to about $2^X(0.8, 0.6)$. So the curve heads up and to the right, roughly in the direction of slope $\frac{3}{4}$.



Part (iii)

The graph $y = \sin x$ for $0 \le x \le 2\pi$ is the familiar single wave.

For $y = \sin(x - 2y)$, put $u = x - 2y$. A point (x, y) is on the curve if and only if $y = \sin u$, and then $x = u + 2y$. So the curve is exactly

$$x = u + 2\sin u, \qquad y = \sin u.$$

This is $y = \sin x$ with every point pushed sideways by twice its height: a horizontal shear $(u, v) \mapsto (u + 2v, v)$.

Which u give $0 \le x \le 2\pi$? Let $x(u) = u + 2\sin u$. Then $x(-u) = -x(u)$ and $x(u + 2\pi) = x(u) + 2\pi$. Since $x'(u) = 1 + 2\cos u$, the function $x(u)$ increases on $[0, \frac{2\pi}{3}]$ from 0 to

$\frac{2\pi}{3} + \sqrt{3} \approx 3.83$, then decreases on $[\frac{2\pi}{3}, \pi]$ down to π. So it lies in $(0, 2\pi)$ for $0 < u \leq \pi$, and on $[\pi, 2\pi]$ the symmetry $x(2\pi - u) = 2\pi - x(u)$ keeps it in $(0, 2\pi]$. For $u > 2\pi$ we get $x(u) = x(u - 2\pi) + 2\pi > 2\pi$ by induction, and for $u < 0$ oddness gives $x(u) < 0$. So the required piece of the curve is traced exactly once as u runs from 0 to 2π, from $(0, 0)$ to $(2\pi, 0)$.

Tangents. Along the curve

$$\frac{\mathrm{d}x}{\mathrm{d}u} = 1 + 2\cos u, \qquad \frac{\mathrm{d}y}{\mathrm{d}u} = \cos u, \qquad \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\cos u}{1 + 2\cos u}.$$

These two derivatives are never zero together, since $\cos u = 0$ makes $1 + 2\cos u = 1$.

- Horizontal tangents occur where $\cos u = 0$, i.e. $u = \frac{\pi}{2}, \frac{3\pi}{2}$:

$$\left(\tfrac{\pi}{2} + 2,\ 1\right) \quad \text{and} \quad \left(\tfrac{3\pi}{2} - 2,\ -1\right).$$

- Vertical tangents occur where $\cos u = -\frac{1}{2}$, i.e. $u = \frac{2\pi}{3}, \frac{4\pi}{3}$:

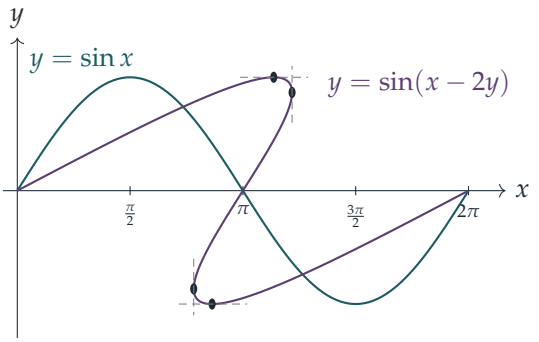
$$\left(\tfrac{2\pi}{3} + \sqrt{3},\ \tfrac{\sqrt{3}}{2}\right) \quad \text{and} \quad \left(\tfrac{4\pi}{3} - \sqrt{3},\ -\tfrac{\sqrt{3}}{2}\right).$$

Numerically these are about $(3.57, 1)$, $(2.71, -1)$, $(3.83, 0.87)$ and $(2.46, -0.87)$.

Ask yourself The horizontal tangent at the top is at $x \approx 3.57$, but the curve still reaches $x \approx 3.83$ after that. What does the curve actually look like?

Follow u from 0 to 2π. The curve leaves the origin with gradient $\frac{1}{3}$ (put $u = 0$), which is shallower than $y = \sin x$. It rises to its maximum at $(\frac{\pi}{2} + 2, 1)$, then moves right and down to the vertical tangent at $(\frac{2\pi}{3} + \sqrt{3}, \frac{\sqrt{3}}{2})$. Then it bends back to the left, passing through $(\pi, 0)$ with gradient $\frac{-1}{-1} = 1$. It reaches the second vertical tangent at $(\frac{4\pi}{3} - \sqrt{3}, -\frac{\sqrt{3}}{2})$, turns right through the minimum $(\frac{3\pi}{2} - 2, -1)$, and rises to $(2\pi, 0)$. The curve overhangs like a breaking wave. Between $x \approx 2.46$ and $x \approx 3.83$ there are three values of y for each x. The whole picture has half-turn symmetry about $(\pi, 0)$.

For interest, the two graphs meet where $\sin x = \sin(x - 2\sin x)$. Either $\sin x = 0$, giving $x = 0, \pi, 2\pi$, or $x - \sin x = \frac{\pi}{2}$ or $\frac{3\pi}{2}$, giving $x \approx 2.31$ and $x \approx 3.97$.



Checks

- Part (i): take the vertex $(3,0)$ of G_1. It maps to $(\frac{3}{\sqrt{2}}, \frac{3}{\sqrt{2}})$, and both brackets in the equation of G_2 give 3 and 0, so the left-hand side is 1.
- Part (ii): $M\binom{1}{2} = \binom{-0.6+1.6}{0.8+1.2} = \binom{1}{2}$, as an invariant point must. The point $(1,2)$ also satisfies the second equation: $0.8 + 1.2 = 2 = 2^{-0.6+1.6}$.
- Part (iii): at $(\frac{\pi}{2}+2, 1)$ we get $x - 2y = \frac{\pi}{2}$, and $\sin\frac{\pi}{2} = 1 = y$. Similarly $(\frac{2\pi}{3} + \sqrt{3}, \frac{\sqrt{3}}{2})$ gives $x - 2y = \frac{2\pi}{3}$, and $\sin\frac{2\pi}{3} = \frac{\sqrt{3}}{2}$. Implicit differentiation gives $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\cos(x-2y)}{1+2\cos(x-2y)}$. Its numerator is 0 at the first point and its denominator is 0 at the second.

Common slips

- Showing only that G_1 maps *into* G_2. To say G_2 *is* the rotated curve you also need every point of G_2 to come from G_1, which is the inverse map.
- Treating $y = \sin(x - 2y)$ as a function of x, so the curve gets drawn as a single wave without its overhang. The vertical tangents are exactly where it folds back.
- Forgetting that the reflected exponential has an oblique asymptote, $4x + 3y = 0$, and lies wholly on one side of it.

Beyond STEP

The principle behind this question is that applying an invertible map T to a curve $f(x,y) = 0$ gives the curve $f\big(T^{-1}(x,y)\big) = 0$. In a first-year linear algebra course this becomes the study of how quadratic forms change under a change of basis. Rotating axes to remove the xy term from a conic is the two-dimensional case of diagonalising a symmetric matrix. Strang's *Introduction to Linear Algebra* covers rotations, reflections and their eigenvectors (the invariant line in (ii)(a) is an eigenvector with eigenvalue 1) very readably.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.